



BrAInVision: A hybrid explainable Artificial Intelligence framework for brain MRI analysis

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ABSTRACT

Brain tumors pose a significant medical challenge, characterized by high incidence and mortality rates, which underscore the critical need for accurate and early diagnosis using minimally invasive techniques such as magnetic resonance imaging. In this context, Artificial Intelligence has emerged as a promising tool to enhance diagnostic precision and efficiency. However, its widespread adoption in clinical practice remains limited due to the opacity of Artificial Intelligence-driven decision-making processes. To address this challenge, we introduce *BrAInVision*, a hybrid and doubly explainable AI framework for brain tumor detection. The novelty of our approach is the integration of both deep learning and traditional machine learning techniques, combining deep-extracted features with hand-crafted features to create a more robust and interpretable classification system. In contrast to conventional single-explanation methods, our framework provides comprehensive explainability through a multi-level analytical approach, enhancing both interpretability and transparency. The first level employs Grad-CAM to visualize regions of interest identified by the deep feature extractor, while the second level utilizes Permutation Feature Importance and Partial Dependence Plots to understand and quantify the contribution of specific image characteristics to diagnostic decisions. The proposed framework achieved an F1-score of 97% on the four classes (Glioma/Meningioma/Pituitary/NoTumor) and an average 99% in binary classification (Glioma/NoTumor), outperforming current state-of-the-art methods. The proposed approach has been validated on both the original dataset and an independent dataset with radiologist-annotated tumor masks, demonstrating strong generalizability. Designed for seamless integration into radiologists' workflows as a decision support system, *BrAInVision* ensures a high degree of explainability, thereby fostering greater trust in AI-assisted medical decision-making.

1. Introduction

The detection and classification of brain tumors remain among the most critical tasks in neuro-radiology, carrying significant implications related to patient survival and quality of life. Globally, approximately 347,992 new cases of brain cancer were recorded in 2019, with 187,491 males and 160,501 females diagnosed [1]. About 24,820 malignant tumors of the brain and spinal cord are expected to be diagnosed in 2025 in the United States only, which can lead to approximately 18,330 deaths [2]. Unfortunately, despite great advances in treatment modalities, the five-year survival rate for malignant brain tumors remains below 36%, clearly evidencing the paramount importance of early and accurate diagnosis. These statistics obviously underline the urgent need for accurate and early diagnosis to improve

patient life expectations. Magnetic resonance imaging (MRI) is the main technique used for the diagnosis and classification of brain tumors. However, interpreting these images requires specialized expertise from radiologists. This is both time-consuming and expensive and can contribute to fatal delays in diagnosis. In recent years, *Artificial Intelligence* (AI) has emerged as a powerful tool in medical diagnostics, its integration into medical frameworks aims to alleviate these challenges.

However, the adoption of AI in clinical practice is not without challenge and faces significant challenges. Some studies [3,4] have shown that most individuals do not fully trust AI-generated medical advice with concerns about the transparency of the AI's decision-making process. Indeed, traditional AI models often operate as “black boxes”, providing almost no insight into their decision-making process [5]. This lack of transparency is a significant challenge and barrier in clinical

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settings nowadays where understanding the reasons behind a diagnosis is crucial for both clinicians and patients. So, it is mandatory to develop *eXplainable AI* (XAI) tools or methodology that provide clear insights into their processes to enlarge trust in AI [6,7].

The understanding of AI decisions is a key factor for clinical adoption [8]; XAI systems receive significantly higher acceptance rates among healthcare professionals compared to non-explainable counterparts [9–11].

Incorporating XAI into brain tumor diagnostics could help to enhance the trustworthiness of AI-assisted decisions, while also could facilitate better communication between clinicians and patients. In addition, regulatory frameworks increasingly require explainability for AI systems, particularly in healthcare settings, making transparency not only a clinical necessity but also a legal and ethical requirements.

This paper has two main objectives: build an AI framework, aiming at the widest possible explainability, and increase the trustworthiness towards healthcare AI. The implementation of two complementary layers of explainability is the main innovation of our approach. The first layer offers visual interpretations of how the deep learning component analyzes MRI images, whereas the second layer provides feature-level explanations of the classification decisions. This strategy performs a dual-explanation approach that addresses the “black box” problem of AI systems from multiple perspectives, and helps in making the framework’s decisions transparent and trustworthy for clinical use. Thereby, we involve both *machine* and *deep learning* pipelines proposing *BrAlnVision*, a *hybrid* and *dually explainable* framework for brain tumor detection. In the proposed solution, two kinds of feature extraction, and consequently two levels of explanation, were implemented: *deep-based* and *hand-crafted*. The deep features are exploited to conduct an image visual analysis; *Grad-CAM* (Gradient-weighted Class Activation Mapping, a technique that highlights regions in images that influence model predictions) is applied to explain what areas are considered relevant by the framework for diagnosis. The hand-crafted features are extracted without the need for a segmentation pipeline and they are merged with deep features to create a new numerical and interpretable dataset. The better-evaluated machine learning model was chosen to produce the final classification. After this step, the second step of interpretation is performed, involving the hand-crafted features employing *Permutation Features importance (PFI)* (PFI, a method that measures feature relevance by evaluating model performance when feature values are randomly shuffled) and *Partial Dependence Plots (PDP)* (PDP, a technique that visualizes the relationship between input features and model predictions). The hybrid approach proposed in this paper addresses the interpretability-accuracy trade-off that typically challenges AI systems in healthcare, achieving both high performance and comprehensive explainability. The proposed framework achieved an *F1-score* of 97% on the Brain Tumor Data dataset (containing four classes: Glioma, Meningioma, Pituitary and NoTumor), outperforming current state-of-the-art methods.

In summary, the main contributions of this work are:

- building a new framework that offers high accuracy while providing *multi-level explainability*, potentially bridging the gap between advanced AI techniques and clinical adoption requirements;
- building an *accurate diagnostic framework* of three distinct types of brain tumors (*glioma*, *meningioma*, *pituitary tumor*) distinguishing them from a *healthy* class;
- introducing a first layer of visual explainability through Grad-CAM that interprets how deep extractor processes MRI images; implementing a second layer of global, model-agnostic explainability at the feature level through PFI and PDP;
- exploring a new medical dataset while validating the framework on a well-known dataset widely used in literature, enhanced with tumor masks validated by radiologists.

The above mentioned contributions address the critical need for transparent AI systems in neuroradiology, where clinical decision-making requires both accuracy and results interpretability. By equipping radiologists with both accurate predictions and clear explanations of those predictions, *BrAlnVision* represents an important step towards AI systems that can be confidently integrated into clinical practice. The paper is organized as follows. Section 2 provides a quick overview of the current state-of-the-art in brain tumor classification using *hand-crafted or hybrid feature extraction techniques* showing the gaps between literature and actual research. In Section 3 conceptual architecture blocks are introduced, while the framework architecture is introduced in 4. A comprehensive analysis of the results is presented in 5 and discussed in 6, analyzing the framework in terms of *performance and explainability*. Finally, limitations and future research directions are discussed.

2. Related works

Numerous studies have explored different approaches to classify tumors in MRI scans enhancing AI and medical image processing. On the other hand, the challenge remains due to solutions interpretability and dataset limitations. This section reviews key contributions in the field highlighting recent advancements and their impact. We organize our review focusing on three main methodological approaches: *deep learning-based techniques*, *textural feature extraction methods* and *explainable AI frameworks*.

2.1. Deep learning approaches

The most widely used approach in the literature concerns feature extraction using stand-alone deep learning algorithms. For example, Montalbo et al. [12] utilize the *Vision Transformers* and *Convolutional Neural Networks*. The authors investigate ViT, MobileViT and Swin Transformers benchmarking them against a *ConvNeXt* CNN model. The dataset consists of 7,023 MRI images with four classes: no tumor, glioma, meningioma and pituitary tumor. Results indicate that transformer-based models outperform CNNs in terms of tumor detection accuracy (95%) but face challenges related to computational complexity and data requirements. An improvement regarding the data privacy is implemented by Albalawi et al. [13] which use a tuned *VGG16* architecture in a federated learning-based architecture. Unlike conventional methods, *federated learning enables decentralized model training* ensuring data privacy across multiple institutions. The model is trained on the Msoud dataset, ensuring robust feature extraction across various tumor types. The approach achieves high precision with an overall accuracy of 98%.

2.2. Textural feature extraction methods

While deep learning approaches show impressive performance, textural feature extraction methods often require fewer computational resources and offer greater interpretability. In this review, we take into account the works that do not use segmentation, therefore the hand-crafted features will refer to the whole magnetic resonance image. Selvy et al. [14] use Histogram Equalization for image enhancement followed by feature extraction using the *Gray Level Co-occurrence Matrix* (GLCM, a statistical method that examines texture by analyzing spatial relationships between pixel intensities). The extracted features are then fed into a *Probabilistic Neural Network* classifier to distinguish between normal and glioma-affected brain MRI scans. The dataset consists of 150 MRI images divided into training and testing sets. The proposed model achieved an accuracy of 90,9%. Kaplan et al. [15] introduce a Pyramid and Fixed-Size Patch approach integrating *Histogram of oriented gradients* (HOG) features and a *k-Nearest Neighbor classifier* (KNN). This study extends beyond brain tumors, incorporating Alzheimer’s Disease classification. The proposed model achieves high accuracy,

about 95% on the first brain tumor dataset, 98,19% on the second brain tumor dataset and 97,80% on a dataset obtained merging the two previous datasets. Balasubramanian et al. [16] proposes a **hybrid deep learning model** that integrates a Random Forest classifier with a Shallow Convolutional Neural Network (CNN). The Random Forest classifier is used for feature engineering, selecting between GLCM, statistical and *Gabor Wavelet* features and ensuring that only the most discriminative features are retained. This hybrid approach aims to improve classification performance while reducing overfitting in limited medical imaging data and achieving a 93,7% of accuracy in the best environment.

2.3. Explainable AI frameworks

As **interpretability becomes increasingly crucial in medical applications**, numerous researchers have intensively worked on developing explainable frameworks. Saeed et al. [17] introduce *Neuro-XAI*, a deep learning framework based on two deep feature extraction modules. The study utilizes pre-trained models with integrated a XAI technique, *Grad-CAM* (which produces visual explanations for CNN decisions by highlighting important regions in the input image), to enhance model interpretability. The experimental results indicate a 97% of accuracy, showing the fact that the deep extraction modules, based on *DarkNet53* and *MobileNet*, are not worthy to classify brain tumor despite the computational effort. Appiah et al. [18] integrates **Proper Orthogonal Decomposition** with deep learning techniques for brain tumor detection. The method extracts essential spatial features before feeding them to a fine-tuned CNN and uses *SHAP* to provide a visual diagnosis. This pre-processing step mitigates the computational complexity. The study evaluates its approach using the BraTS 2016 dataset and achieves a 92,78% in classification accuracy, surpassing traditional deep learning models that rely solely on raw MRI images.

2.4. Research gaps and our contribution

Based on the performed literature review, we have identified several significant gaps in the current state-of-the-art. First, there is limited use of **hand-crafted features** in brain tumor classification, and when employed, they are rarely utilized for enhancing the interpretability of the diagnostic process. Second, deep features though extensively employed, are often underutilized, particularly when accuracy is low, when only binary classification is performed, or when it is not applied to XAI methods. Third, most existing approaches just implement single-level explanations, which offer limited insight into the model's decision-making process.

Initially, our study aimed to enhance the existing literature using the widely adopted dataset [19], improving it with the *accuracy* of the deep features, the *interpretability* of hand-crafted features and incorporating visual explainability techniques. However, we decided to use a recently published medical dataset on Mendeley Data [20] to train and test the framework. Furthermore, the proposed methodology was validated using the first dataset enhanced with radiologist-validated segmentation [21] to ensure a rigorous analysis. Although the dataset used to train the model was introduced in early 2024, so the amount of existing literature is scarce, consisting of only two studies.

The first one by Mohan and Ramkumar [22] used MobileNet, a CNN architecture optimized for devices with limited computational resources. The research begins by cutting the Mendeley Brain Tumor Dataset and consequently considering 600 MRI images divided equally between glioma tumor cases and non-tumor cases. A data augmentation technique was applied to artificially expand the dataset, which includes rotation (20 degrees), scaling, shear transformation and horizontal/vertical flips. Additional layers were introduced, including average pooling, batch normalization and fully connected dense layers. The final layers of the network consisted of fully connected neurons with 2048, 1024, 256, and 128 units with dropout layers (0.5

Table 1

State of the art approaches on Mendeley brain tumor dataset.

Ref	Methodology	Results (Accuracy/F1-Score)	Considerations
[22]	Deep classification only with MobileNet	93%/93%	No ML models no hand-crafted features no XAI or interpretations binary classification about 50 test images
[23]	Deep classification only with InceptionV3	98%/98%	No ML models no hand-crafted features no XAI or interpretations binary classification about 50 test images

dropout rate). The Adam optimizer was used, with a learning rate of 0.0001, a batch size of 34 and a total of 50 training epochs. The results recorded an accuracy of 93,33%, a sensitivity of 88,89%, a specificity of 96,97% and an F1-Score of 93%. Additionally, a confusion matrix was generated to visualize the model's classification errors on a test set of about 50 MRI images. The second work is also written by Mohan and Ramkumar [23] and proposed an InceptionV3-based model. The methodology is substantially the same as the previous work. The algorithm achieved an accuracy of 98,33%. Despite the performance of these works, the classification they propose is binary (*Glioma/NoTumor*) and the numerosity of the test set is very low as stated in Table 1.

Based on previous described gaps, our work provides several significant contributions. We propose a hybrid framework that integrates deep features with hand-crafted features with the final goal of achieving both high accuracy and interpretability. On the contrary of existing approaches that offer only single-level explanations, our framework provides dual-level explainability through both visual (*Grad-CAM*) and feature-level (PFI and PDP) interpretations. Furthermore, we perform multi-class classification on larger test sets and ensure robustness and generalizability by validating our approach across different datasets.

3. Background

This section provides an overview of the fundamental techniques and algorithms that form the backbone components of our brain tumor classification framework (*BraInVision*). Understanding these building blocks is essential for appreciating the hybrid and explainable approach presented in this work.

3.1. VGG16

VGG16 [24] is a deep CNN architecture developed by the **Visual Geometry Group (VGG)** at the University of Oxford. The name VGG16 refers to the 16 weight layers which consist of 13 convolutional layers and 3 fully connected layers. The convolutional layers use small 3×3 filters with a stride of 1 and padding to maintain spatial dimensions. These layers extract features from the input image, followed by a **Rectified Linear Unit activation function** that introduces non-linearity. Additionally, max pooling layers with 2×2 filters and a stride of 2 are used to downsample feature maps and reduce computational complexity. After the convolutional and pooling layers, the network includes three fully connected layers. The first two fully connected layers contain 4096 neurons each, while the final output layer contains neurons corresponding to the number of classes in the dataset or the number of features extracted. Feature extraction in VGG16 occurs through convolutional layers, which apply filters to detect edges, textures and complex patterns across different depths of the network. One of the advantages of VGG16 is its simple and uniform architecture, making it easy to implement and adapt for **transfer learning applications in medical imaging**.

3.2. Machine learning models

Our framework employs several machine learning algorithms for final classification task based on the extracted features:

- **Support Vector Machines (SVM)** [25]. SVM is a supervised learning algorithm used for classification and regression that finds the optimal hyperplane that separates data points belonging to different classes in a high-dimensional space. The optimal hyperplane is determined by maximizing the margin, which is the distance between the nearest data points (support vectors) from each class to enhance generalization. Common kernel functions include linear, polynomial, and radial basis function (RBF), each suited for different types of data structures. Despite its ability to rigorously manage large dimension dataset, SVM can be computationally intensive and requires careful selection of parameters to achieve optimal performance.
- **Random Forest Classifier (RF)** [26]. RF is an ensemble learning method primarily used for classification and regression tasks which operates by constructing multiple decision trees during training and aggregating their predictions to improve accuracy and reduce overfitting. The algorithm is an extension of bagging (Bootstrap Aggregating), where each decision tree is trained on a random subset of the data sampled with replacement. The effectiveness of Random Forest is attributed to its ability to handle high-dimensional datasets, robustness to noise, but can be computationally expensive, particularly with large datasets.
- **K-nearest neighbors (KNN)** [27]. KNN is a simple yet powerful supervised learning algorithm used for classification and regression tasks. Unlike many machine learning models, KNN does not build an explicit mathematical model during training, but instead perform a “lazy” learning, where it simply stores the training data and makes predictions by comparing new data points to existing ones. While KNN may be a lightweight and easy-to-implement solution, it can be computationally expensive for large datasets since every prediction requires measuring distances from all training points. Additionally, KNN is sensitive to irrelevant or highly imbalanced features, making feature scaling an important preprocessing step.
- **Extreme Gradient Boosting (XGBoost)** [28]. XGBoost is a highly efficient and scalable machine learning algorithm based on gradient boosting, an ensemble learning technique. It is widely used for both classification and regression tasks, particularly in structured tabular data. The core idea of XGBoost is to build multiple decision trees sequentially, where each new tree aims to correct the errors made by the previous ones. It employs several optimization techniques, such as tree pruning, parallel processing, and regularization, to reduce overfitting and improve speed and it also supports handling missing values automatically, making it highly robust in real-world scenarios, even when they are described by complex patterns.

3.3. Grad-CAM

Grad-CAM (Gradient-weighted Class Activation Mapping) is a technique used to visualize the regions of an image that are important for a CNN in a classification task [29]. This technique is central to our framework's first layer of explainability, providing visual insights into the deep extractor focus area. Grad-CAM is largely used for interpreting CNN results, increasing explainability and improving debugging processes. The algorithm operates as follows: firstly, the network produces class scores through its layers for a given input image, then the gradients of the target class score are computed with respect to the feature maps of a convolutional layer via backpropagation. These gradients are then averaged over width and height dimensions to obtain importance weights for each feature map channel. These weights highlight the

significance of each feature map for the target class and they are used to perform a weighted combination of the feature maps from the convolutional layer. The result is a coarse localization map highlighting the important regions of the image. This map is then passed through a ReLU function to ensure only positive influences on the class score are considered. The final class activation map is upsampled to the size of the input image and superimposed on the original one. The output is a heatmap showing the image regions most influential to the model's prediction, providing radiologists with visual confirmation of the areas influencing the diagnostic decision.

3.4. Permutation Feature Importance

Permutation Feature Importance (PFI) is essential to our framework's second layer of explainability, revealing which features most significantly influence classification decisions. The importance of a feature is determined by measuring the increase in the model's prediction accuracy after permuting its values. A feature is considered *important* if shuffling its values results in a significant increase in model error, indicating that the model relied on that feature for prediction. Conversely, a feature is deemed *unimportant* if permuting its values has no impact on the model error, implying that the model did not utilize that feature for prediction. PFI metric was introduced by Breiman [26] for its random forests model. Expanding on this idea, Fisher et al. [30] proposed a *model-agnostic* version, referred to as *model reliance*. Following Fisher's approach, let $\hat{f}$ be the trained model, X the feature matrix, y be the target vector, and $L(y, \hat{f})$ the error measure.

Permutation feature importance offers several advantages. One of its key aspects is interpretability because the *importance measure* is directly related to the increase in model error (or accuracy, in our case). This makes easy to understand the *impact of each feature on model performance*. In this sense, it provides global insight of the model's behavior, helping to identify which features contribute the most to predictions. Comparability is another key aspect since permutation feature importance relies on an error ratio rather than an absolute error difference. Another benefit is that *no retraining is required*. Unlike methods that remove features and retrain the model, permutation feature importance evaluates features within a fixed model [31].

3.5. Partial Dependence Plot

Partial Dependence Plot (PDP) complements PFI in our framework by visualizing how specific features affect the model's predictions, further enhancing the interpretability of our classification results. PDP is a *global and model-agnostic XAI technique* [32]. This method provides a comprehensive interpretation by analyzing the influence of one or more features on the target variable across all observations. In contrast, local interpretability techniques focus on assessing the impact of individual features at the observation level. The model-agnostic nature of PDP allows it to be applied to any predictive model, regardless of its underlying architecture. PDP allow us to examine the *marginal effect* of specific features on the model's predictions, thereby offering insights into their contribution to the overall decision-making process of the black-box model.

4. Architecture

BraInVision stems from the critical need to develop an AI framework that ensures high diagnostic accuracy and transparent decision-making in the challenging domain of brain tumor diagnostics. Our approach embraces *four conceptual phases*, each designed to progressively transform raw images into a clinically relevant decision:

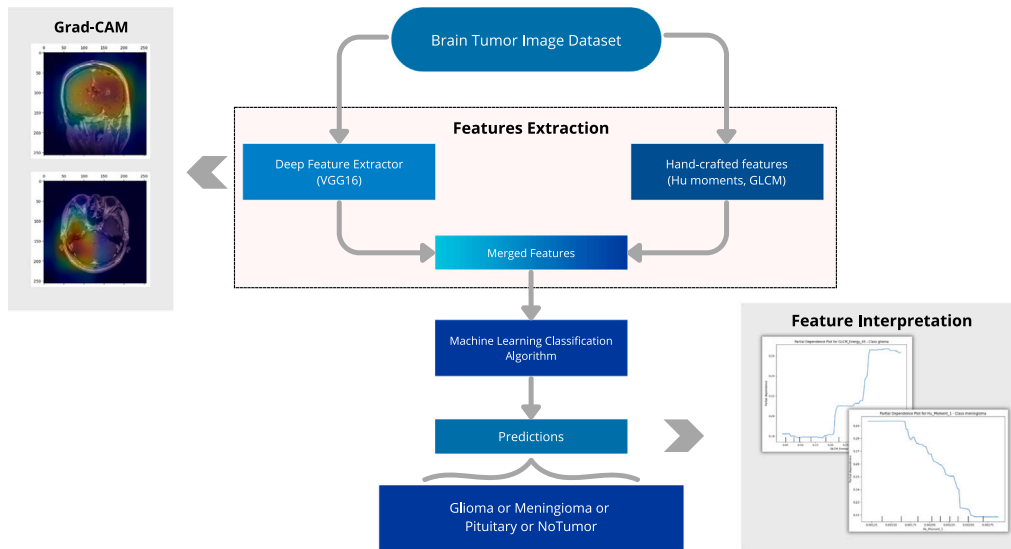


Fig. 1. *BrAlnVision* framework overview.

- **Data Preprocessing:** this initial phase, handles MRI images as raw material to be refined. More in details, images belonging to four classes (*Glioma*, *Meningioma*, *Pituitary tumor*, or *NoTumor*) are loaded and preprocessed to standardize their format and dimensions. Image normalization, resizing, and color space transformations are applied in order to guarantee consistent input representation regardless of acquisition parameters. The aim of this phase is to create a clean, standardized dataset ready for analysis.
- **Deep Feature Extraction:** deep features are extracted using a custom deep features extractor based on an *VGG16* network to capture complex visual patterns which are not immediately visible in the images. This phase applies transfer learning by utilizing pre-trained *ImageNet* weights, allowing the network to extract rich hierarchical features despite limited domain-specific training data.
- **Feature Engineering:** this phase represents the core of our hybrid approach. *BrAlnVision* involves not only features extracted by the neural network in the precedent step, but enrich them with additional hand-crafted features, derived from *Hu moments* and *GLCM*. These traditional features are extracted and merged with deep features in order to obtain a complex representation vector for each original image. These combined features are normalized using *Z-Score Standardization* [33]. Above the technical details, the merging of these two features types intuitively ensure the dialogue between two relevant domain nuances: the computational intuition encoded in features extracted by deep neural networks and the soundness of traditional clinical analysis encoded in hand-crafted features.
- **Classification:** in the final phase, a machine learning model performs the final classification, receiving the combined features array, and providing a clinical decision, corresponding to one of the four tumor categories. In the proposed architecture, we deliberately separate feature extraction from classification to enhance explainability at multiple levels of the framework.

BrAlnVision implements a hybrid approach that bridges the gap between deep learning's predictive power and traditional machine learning's interpretability, therefore offering a unique solution to medical image analysis challenges. More specifically, our framework addresses the crucial requirement for explainability in medical AI systems by incorporating different explanation techniques at multiple levels of the classification pipeline:

- *Grad-CAM* is used in the first framework phase to visualize how the deep extractor focused on images and then extracted the most important features, making the extraction process clearer.
- *PFI* and *PDP* are used in the final framework phase to fully understand how hand-crafted features contributed in the classification process, firstly assessing the impact of each feature on the model's prediction and then analyzing the relationship between specific features and the model's output. These techniques guarantees transparency as they provide quantitative evidence of feature relevance and their influence on specific diagnostic decisions.

The proposed hybrid approach enhances accuracy while ensuring explainability, making the solution's insights more accessible to domain experts and facilitating practical clinical applications.

Our architectural design supports radiological expertise as it prioritizes both diagnostic accuracy and clinical workflow integration by offering visual analysis alongside feature-level explanations. A detailed description of used hand-crafted features is proposed in the next subsection of this work. The proposed hybrid explainable framework, *BrAlnVision*, is reported in Fig. 1, while the methodology phases are formalized in Algorithm 1.

Algorithm 1 formalizes the training workflow of our framework, illustrating the sequential processing steps from image preprocessing through feature extraction and classification. The algorithm illustrates how deep and hand-crafted features are merged to create a unified representation that balances predictive power with interpretability.

4.1. Hand-crafted features

Hand-crafted features are a critical component of our explainable framework as they provide objective evidence, such as domain specific measurement, that are inherently interpretable to clinical experts. In the following section, we formally introduce the hand-crafted features used in the classification process: *Hu Moments* [34] and *GLCM* [35]. The idea behind the combination of these features was twofold: providing a system that employs reproducible techniques which are interpretable and model-agnostic, while leveraging their complementary properties in characterizing both the morphological aspects (shape and geometry) and textural patterns of brain tissues, which are known to be particularly relevant for distinguishing between different types of brain tumors.

Algorithm 1 BrAIIn Vision training

Input: Dataset $D_{\text{train}}, D_{\text{test}} = (x_i, y_i)_{i=1}^N$, where x_i are images, $y_i \in \{0, 1, 2, 3\}$ are tumor labels and n is the number of instances

Output: Function $f : \mathcal{X} \rightarrow \{0, 1, 2, 3\}$, where $\mathcal{X}$ is the sample space

- 1: **Data Preprocessing:**
- 2: $D_{\text{train}}, D_{\text{test}} \leftarrow \text{PreprocessImages}(D_{\text{train}}, D_{\text{test}})$ $\triangleright$ Load images and resize
- 3: **Deep Feature Extraction:**
- 4: Initialize pre-trained VGG16Net M_{CNN}
- 5: $F_{\text{deep}} \leftarrow \text{ExtractDeepFeatures}(D_{\text{train}})$ $\triangleright$ 256 deep features
- 6: **Feature Engineering:**
- 7: $F_{\text{hand}} \leftarrow \text{ExtractStatisticalFeatures}(D_{\text{train}})$ $\triangleright$ 27 hand-crafted features
- 8: $F_{\text{combined}} \leftarrow [F_{\text{deep}}; F_{\text{hand}}]$ $\triangleright$ Concatenate features
- 9: **Classification:**
- 10: **for** $k \in 1, 2, 3, 4$ **do**
- 11: Train ML model f_k on $(F_{\text{combined}}, D_{\text{train}}[1])$ $\triangleright D_{\text{train}}[1]$: labels in D_{train}
- 12: $y_{\text{pred}}^k \leftarrow f_k(F_{\text{combined}})$ $\triangleright$ ML model predictions
- 13: **end for**

4.1.1. Hu moments

The Hu moments, introduced by Hu [34] in 1962, are a set of seven *invariant* moments derived from *central moments* of an image. These moments are particularly useful in pattern recognition and shape analysis [36–38] as they remain invariant under *translation*, *scaling* and *rotation*. It is important to point out that, in the specific domain of brain MRI analysis, these invariant properties are particularly relevant as they allow for consistent feature extraction regardless of tumor orientation or positioning within the scan while maintaining their model-agnostic nature. The central moments of an image $I(x, y)$ are defined as:

$$\mu_{pq} = \sum_x \sum_y (x - \bar{x})^p (y - \bar{y})^q I(x, y) \quad (1)$$

where $\bar{x}$ and $\bar{y}$ are the centroid coordinates of the image. From these central moments, Hu defined seven invariant moments:

$$\phi_1 = \eta_{20} + \eta_{02} \quad (2)$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2 \quad (3)$$

$$\phi_3 = (\eta_{30} - 3\eta_{12})^2 + (3\eta_{21} - \eta_{03})^2 \quad (4)$$

$$\phi_4 = (\eta_{30} + \eta_{12})^2 + (\eta_{21} + \eta_{03})^2 \quad (5)$$

$$\phi_5 = (\eta_{30} - 3\eta_{12})(\eta_{30} + \eta_{12}) [(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] + (3\eta_{21} - \eta_{03})(\eta_{21} + \eta_{03}) [3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \quad (6)$$

$$\phi_6 = (\eta_{20} - \eta_{02})(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2 + 4\eta_{11}(\eta_{30} + \eta_{12})(\eta_{21} + \eta_{03}) \quad (7)$$

$$\phi_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12}) [(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] - (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03}) [3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \quad (8)$$

These seven moments provide a robust description of shape characteristics in images, facilitating object recognition. The first moment ϕ_1 corresponds to the moment of inertia around the image centroid, while ϕ_2 describes variance-related properties such as elongation and symmetry. The higher-order moments (ϕ_3 through ϕ_7) capture increasingly complex shape characteristics that are particularly valuable for differentiating tumor morphologies with subtle differences. Notably, the first six moments are invariant with respect to reflection, while ϕ_7 provides the additional capability to distinguish between mirrored images, which can be critical when assessing anatomical structures in medical imaging.

4.1.2. Gray level co-occurrence matrix features

GLCM is a statistical method used to examine the texture of an image considering the spatial relationship between pixel intensities. Introduced by Haralick et al. [35] in '73, GLCM began a widely used brain tumor characterization method [39–41]. GLCM features, in our proposal, complement the shape-based information provided by Hu moments as they capture the textural heterogeneity characteristic of different tumor types. The GLCM is constructed by computing how often a pixel with a specific gray-level value occurs in relation to another pixel at a defined spatial offset (d, θ) . The matrix is defined as:

$$P(i, j|d, \theta) = \text{number of occurrences} \quad (9)$$

where (d_x, d_y) are the pixel displacements determined by the chosen angle θ , $I(x, y) = i$ and $I(x + d_x, y + d_y) = j$.

The statistical measures derived from the GLCM to characterize texture are:

- **Contrast:** measures the intensity difference between pixels and their neighbors.

$$\text{Contrast} = \sum_{i,j} P(i, j)(i - j)^2 \quad (10)$$

- **Dissimilarity:** quantifies the variation in intensity between neighboring pixels. It is mathematically defined as:

$$\text{Dissimilarity} = \sum_i \sum_j P(i, j)|i - j| \quad (11)$$

- **Correlation:** measures the linear dependency of gray levels.

$$\text{Correlation} = \sum_{i,j} \frac{(i - \mu_i)(j - \mu_j)P(i, j)}{\sigma_i \sigma_j} \quad (12)$$

- **Energy:** also known as Angular Second Moment, represents textural uniformity.

$$\text{Energy} = \sum_{i,j} P(i, j)^2 \quad (13)$$

- **Homogeneity:** measures the closeness of elements in the GLCM to the diagonal.

$$\text{Homogeneity} = \sum_{i,j} \frac{P(i, j)}{1 + |i - j|} \quad (14)$$

Clinically, these features capture important tissue properties: contrast and dissimilarity reflect tissue heterogeneity often associated with malignancy, correlation indicates tumor-specific patterns, while energy and homogeneity quantify tissue uniformity differences between tumor grades.

In this work, we tuned the GLCM matrix to extract the image features from four different angles (0, $\pi/2$, π , $3/2\pi$ radians). This multi-directional approach ensures that textural patterns are captured regardless of their orientation. The GLCM features amount to 5 for each angle for a total of 20 and, combined with the seven Hu moments, bring the total hand-crafted features to 27. It is worth noticing that the computational efficiency of extracting these features is considerably high in comparison to more complex approaches, making them particularly suitable for applications, such as those in the clinical domain, where processing time may be a challenging factor. Additionally, the features interpretability allows clinicians to be aware of which specific aspects of tumor morphology and texture are driving the classification decisions, thereby increasing trust in the system's outputs.

5. Experimental settings and results

This section reports the evaluation of our BrAIInVision framework, detailing the experimental settings, datasets, and performance metrics. We conducted a comprehensive series of experiments using the Google

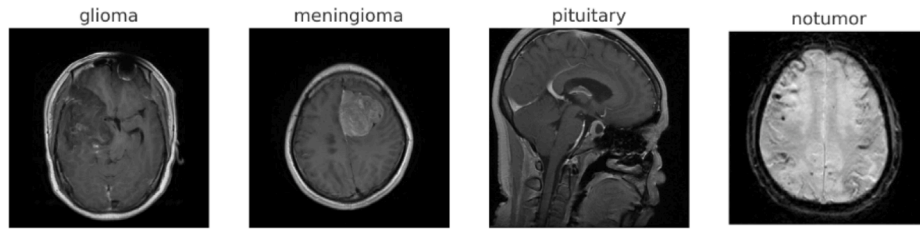


Fig. 2. Images of the dataset.

Colab [42] platform. The T4 GPU runtime and high RAM were employed specifically for deep feature extraction while the remaining processes were executed on standard computational resources. This experimental setup ensures reproducibility while balancing computational efficiency with performance requirements.

5.1. Datasets

For this study, we carefully selected two complementary brain MRI datasets to rigorously evaluate our framework's performance and generalizability. The first dataset served as our primary training and testing resource, while the second dataset provided additional validation with expert-annotated tumor masks. The first dataset, namely *Brain Tumor Data* dataset [20], is a rigorously curated collection of grayscale MRI images of brain tumors organized in two main sets: one dedicated to training and the other for independent testing. Each set is divided into four categories, corresponding to the different types of tumors: *glioma*, *meningioma*, *pituitary* and a further class called *no tumor* which includes images of healthy brains. (see Fig. 2).

For the first experimental evaluation, the training section of the dataset contains 1.321 images of glioma, 1.339 of meningioma, 1.457 of pituitary and 1.595 images of no-tumor brains, making a total of 5.712 images. The testing part, designed to provide an unbiased assessment of model performance, includes 300 images of glioma, 306 of meningioma, 300 of pituitary and 405 of healthy brains which brings the total number to 1.311 images. Overall, the dataset comprises 7.023 images. The dataset exhibits a well-balanced distribution across the different classes, (approximately 25% per class), which is particularly important for avoiding class bias during training and ensures that the model learns to discriminate equally well between all tumor types.

The second dataset, namely *Brain Tumor Segmentation Dataset*, is a well-known selection of brain tumors grayscale MRI images [19] segmented afterwards by radiologists [21]. Both images and masks folders are divided into the four upcited categories, *glioma*, *meningioma*, *pituitary* and *no tumor*. The second experimental evaluation consists in testing the framework on the *Brain Tumor Segmentation Dataset*. This cross-dataset validation approach provides a more stringent test of the framework's robustness and generalizability beyond the training distribution. This type of testing is made to provide an additional proof of the model's robustness. The dataset includes 649 glioma images, 999 of meningioma, 994 of pituitary, which are all provided with tumor masks, and 1.595 of healthy brains where masks are null. The dataset exhibits balanced distribution across the different classes.

All the images are in *JPG* format and have heterogeneous resolutions. Furthermore, the images were acquired using different scan planes, regardless of the class. It is worth noting that some images, although they are grayscale, are encoded as *RGB* (3-channels). This variability in image characteristics reflects real-world clinical scenarios, where standardization of parameters acquisition is often challenging. As part of our preprocessing pipeline, all images are resized to 256×256 pixels and rescaled to ensure consistent input dimensionality across the framework. In addition, they were all uploaded using *RGB* encoding, which triplicates the single grayscale channel, in cases where the three channels do not exist, to facilitate feature extraction using deep learning techniques. It is important to note that from this point

Table 2

Deep learning features extractor summary.

Layer	Hyperparameter
VGG16	<i>weights</i> = 'imagenet', <i>include_top</i> =False
GlobalAveragePooling2D	
Dense	<i>Neurons</i> = 256, <i>activation</i> = 'relu', <i>name</i> = 'features_layer'

onward, the first dataset will be referred to as the original dataset, to imply that it is the dataset on which the model was trained, in accordance with the original sub-division into training and test sets, and the second dataset will be referred to as the validation dataset, since was used entirely as a test set for the classification phases and to assess the quality of the heatmaps by making use of the masks it provides.

5.2. Deep learning feature extractor settings

The choice of using *VGG16* as the deep feature extraction net stems from the results obtained from recent works [43–46], where researchers have compared various state-of-the-art pre-trained networks in brain tumor classification task. These comparative studies showed that *VGG16* consistently outperforms other architectures in terms of Accuracy and F1-score for this specific domain, despite newer architectures often showing superior performance in general computer vision tasks.

Our custom deep features extractor is constructed using a *VGG16* network as base model, leveraging the imagenet pre-trained weights through transfer learning technique without fine-tuning the original architecture. The output of the *VGG16* base model is passed to a *GlobalAveragePooling2D* type level, which effectively reduces the spatial dimensions (height and width) to a single value per channel. Subsequently, a *Dense* layer with 256 neurons processes the output and performs the final feature selection, using a *ReLU* activation function. To facilitate access to the extracted features, this layer was specifically labeled with the name "features_layer". This architectural design balances network depth and feature dimensionality, producing a rich yet manageable set of 256 deep features that capture complex MRI visual patterns.

Table 2 provides a concise overview of the deep feature extractor architecture.

5.3. Machine learning model settings

To identify the optimal classifier for our hybrid framework, we conducted a systematic comparison of four machine learning algorithms [45,47], each selected for its unique strengths and well-established performance in medical classification tasks. Each model offers a different approach to classification, from the instance-based reasoning of *KNN* to the ensemble decision trees of *Random Forest* and *XGBoost*, providing complementary strengths in handling the complex feature space. The best set of hyperparameters for each tested machine learning model has been identified through a systematic approach using the

Table 3
Machine learning hyperparameters.

Model	Hyperparameters
KNN	algorithm = 'auto', leaf size = 1, n neighbors = 1, p = 1, weights = 'uniform'
RandomForest	criterion = 'entropy', max depth = 30, max features = 30, n estimators = 100
SVM	kernel = 'rbf', C = 1
XGBoost	gamma = 0, n_estimators = 100, learning_rate = 0.1

Table 4
5-fold cross validation metrics statistics.

Model	Metric	Min	Max	Mean	Std
KNN	Accuracy	93%	95%	95%	1%
	Recall	93%	95%	94%	1%
	Precision	93%	95%	94%	1%
	F1	93%	95%	94%	1%
RandomForest	Accuracy	91%	93%	92%	1%
	Recall	91%	92%	92%	1%
	Precision	91%	92%	92%	1%
	F1	91%	92%	92%	1%
SVM	Accuracy	91%	94%	93%	1%
	Recall	91%	93%	93%	1%
	Precision	91%	94%	93%	1%
	F1	91%	94%	93%	1%
XGBoost	Accuracy	92%	93%	93%	0%
	Recall	92%	93%	92%	0%
	Precision	92%	93%	93%	0%
	F1	92%	93%	92%	0%

GridSearchCV function, available in the *sklearn* library. The selection was performed by validating parameters through the K-Fold Cross Validation technique, with a K value equal to 5, which provides a robust estimate of model performance while balancing computational costs. This rigorous hyperparameter optimization process ensures that each model reaches its optimal performance, enabling a fair comparison between different classification approaches. Prior to the comprehensive grid search, the models were initially tested by varying different subsets of hyperparameters and evaluating the general performances using default hyperparameters provided by the respective libraries.

Table 3 shows the optimized hyperparameters used for each best-performing model configuration, while Table 4 presents the K-Fold statistical results including accuracy, precision, recall, and F1-score metrics across all validation folds. The comprehensive evaluation across multiple metrics provides a holistic assessment of model performance beyond simple accuracy, ensuring that the selected classifier delivers balanced performance across all tumor classes.

The cross-validation results show strong performance across all models, with KNN achieving the highest scores in all metrics. Low standard deviations (1% or less) indicate stable performance and good generalization. These results guided our final model selection.

6. Discussion

This section analyzes the performance and explainability aspects of our BrAlnVision framework, contextualizing the results within the current state of the art and examining its clinical implications. We present a detailed evaluation across multiple experimental scenarios, highlighting both the quantitative performance and the qualitative interpretability benefits. The proposed explainable framework demonstrated highly encouraging results across multiple evaluation scenarios. Our experimental approach followed a progressive evaluation strategy, beginning with a binary classification task to establish baseline performance and comparability with existing literature, following with the more challenging multi-class classification scenario that represents real-world clinical requirements.

Table 5
Performance comparison.

References	Authors	Accuracy	Precision	Recall	F1-score
[22]	Mohan et al.	93%	91%	97%	93%
[23]	Mohan et al.	98%	100%	97%	98%
BrAlnVision	Gagliardi et al.	99%	99%	99%	99%

Table 6
Models performance.

Model	Accuracy	Precision	Recall	F1-score
KNN	97%	98%	97%	97%
RandomForest	91%	91%	91%	91%
SVM	93%	93%	93%	93%
XGBoost	92%	92%	92%	92%

6.1. Binary classification performance

In order to validate the feature extraction and classification steps proposed, we first conducted an initial dichotomous test focusing only the “Glioma” and “NoTumor” classes, following the evaluation protocol established in [22,23]. Our approach significantly outperformed existing state-of-the-art solutions in the literature, thus effectively validating the proposed architecture. This binary classification scenario provides a direct comparison point with previous work while establishing a performance foundation for our more comprehensive multi-class analysis. Table 5 presents a comparative analysis of the results obtained on the test set, evaluated against previous works that conducted the same test, while Fig. 3 provides a visual representation of model performance through two different confusion matrix, the first one obtained with the original binarized test set (Fig. 3(a)), the second one obtained testing the model on the binarized validation dataset (Fig. 3(b)).

The proposed solution achieves better classification accuracy between the two classes, with zero classification errors, being evaluated on a larger and more challenging test sample than those presented in [22,23]. The high result achieved in the binary case demonstrates the effectiveness of our hybrid feature approach, with the complementary nature of deep and hand-crafted features contributing to the framework's robust discriminative capabilities.

6.2. Multi-class classification performance

Building on the strong binary classification results, we extended our evaluation to the more challenging multi-class scenario involving all four diagnostic categories. The framework maintained robust results regardless of the final classifier used, as reported in Table 6. Among the evaluated classifiers, KNN provided the best classification performance, mirroring its superiority in the dichotomous case, achieving an F1-Score equal to 97% using the original test set, with its detailed performance visualization shown in Fig. 4.

Observing the confusion matrix (Fig. 4(a)), it becomes evident that the framework correctly classifies the great majority of the test data, making only one major error by classifying a meningioma as *NoTumor*. Additionally, it occasionally wrongly classifies meningiomas as gliomas, a challenging distinction even for expert radiologists, due to their similar characteristics. The quality of the proposed framework is also highlighted by the performance achieved on the validation dataset (Fig. 4(b)), where the model obtained an average F1-Score of 99% across the various experimental scenarios. This cross-dataset performance is particularly important, as it testifies the framework's ability to generalize beyond its training distribution to new data sources with different acquisition parameters and patient populations. The effectiveness of the adopted methodology is twofold, as the use of combined features has improved performance from a diagnostic point of view and increased the explainability of the results by introducing features typically used in medical imaging protocols. In Table 7, a

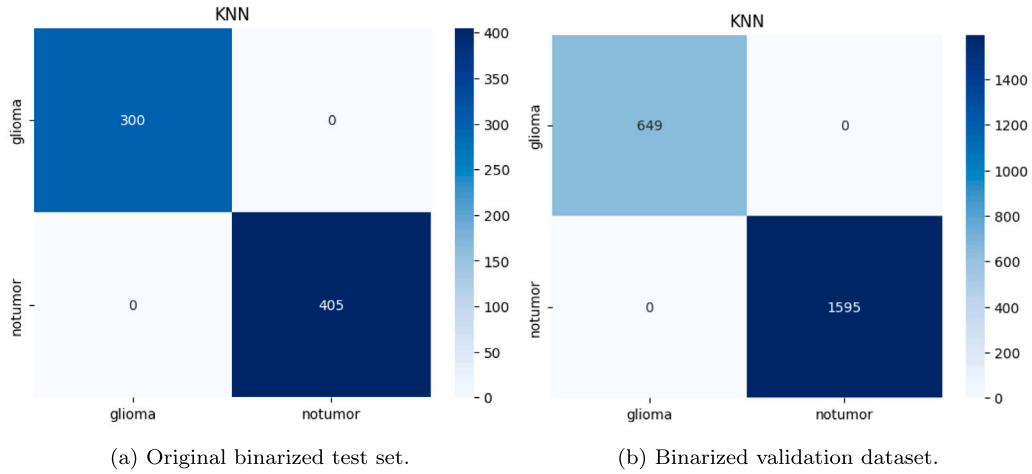


Fig. 3. Confusion matrix related to dichotomy study (Glioma/NoTumor).

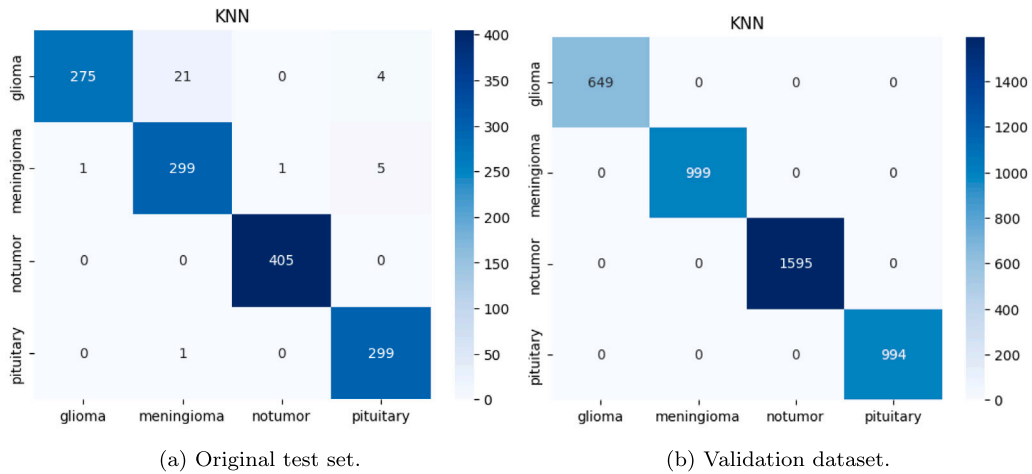


Fig. 4. Final confusion matrix using a KNN classifier.

detailed *ablation study* is provided regarding the performance of the models obtained by varying the set of features used for training and testing. The performance of each combination was validated through 5-fold cross-validation and the results show the performance obtained on the test set. This systematic ablation analysis isolates the contribution of each feature type to overall performance, providing insights into their complementary nature and validating our hybrid approach. The table shows that the KNN model consistently outperformed the others. Deep features alone achieved significant results, which were further improved by combining them with hand-crafted features, while the performance obtained using only hand-crafted features was low but still reliable, likely due to their limited number. This confirms our hypothesis that the integration of both feature types leverages their complementary strengths.

6.3. Framework explainability analysis

This section examines both the visual and feature-level interpretability aspects of *BrAInVision*. Concerning the result explainability, the core contribution of our research, a two-level analysis was conducted: a visual analysis, related to the deep extractor, and a statistical analysis, related to the final classification step. The visual analysis, conducted using the Grad-CAM technique, was designed to highlight the areas of interest identified by the deep feature extractor in the original images. This phase yields a dual outcome: from a technical perspective, it

Table 7

Ablation study altering the features set.

Features configuration	Model	F1-score	Findings
Deep features only	KNN	95%	Baseline results
	RandomForest	90%	
	SVM	91%	
	XGBoost	90%	
Hand-crafted features only	KNN	84%	Performance dropped
	RandomForest	72%	
	SVM	72%	
	XGBoost	73%	
Combined features	KNN	97%	Performance increased
	RandomForest	91%	
	SVM	93%	
	XGBoost	92%	

allows researchers to verify if the extractor is operating rigorously by analyzing the heatmaps produced by the applied technique; from a clinical perspective, it provides a diagnostic contribution, as the framework not only provides a final classification but also highlights specific anatomical regions of interest that determined the output, facilitating expert verification of the system's reasoning.

Fig. 5 presents illustrative examples for each class, demonstrating how the extractor focuses on specific areas of the images to determine the tumor type. Notably, the highlighted areas coincide with regions

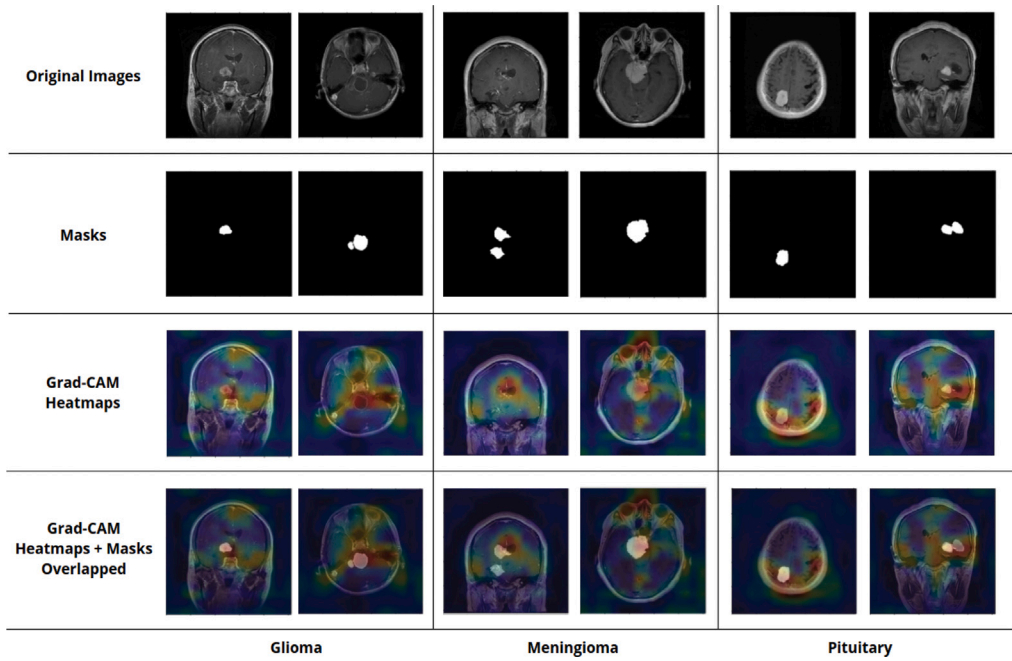


Fig. 5. Grad-CAM visual analysis.

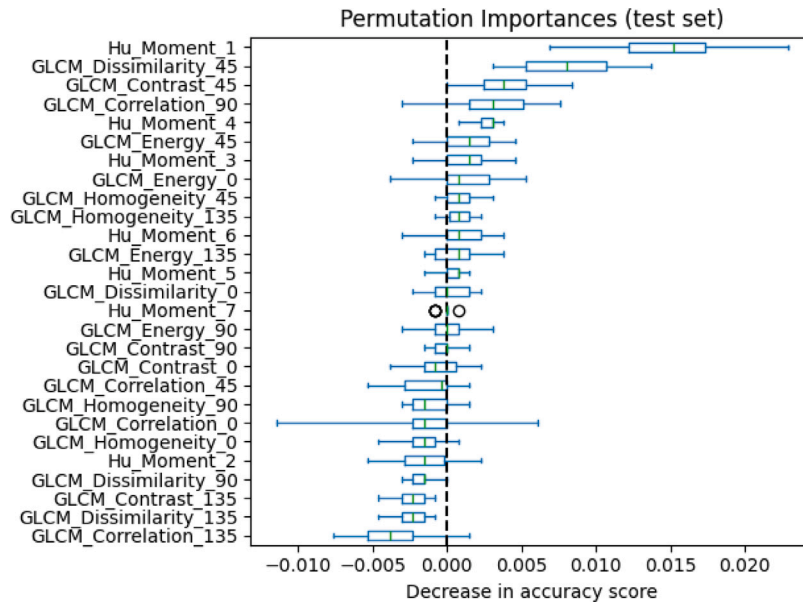


Fig. 6. Hand-crafted features importance.

where cancerous formations of the examined type are predominantly found according to clinical literature, thus providing radiological validation of the system's attention mechanisms. This alignment between the model's areas of focus and clinically relevant regions is crucial for building trust among medical practitioners.

In general, the model achieves rigorous results, classifying the tumors correctly. However, an in-depth analysis of the VGG16 features reveals that the architecture has some limitations focusing the tumor regions. The validation dataset, whose masks were validated by three radiologists [21], was used by overlapping the provided masks with the obtained Grad-CAM to validate the model behavior. For a large portion of the dataset, there is a clear overlap regardless of class, although there are images in which the network does not focus accurately. In fact, using classical overlap evaluation metrics does not yield very high performance. Although the results are still valid, since the model, while

focusing on tumor areas, also tends to include surrounding regions, thus creating a sort of global context, as it was not directly trained on this specific domain. It is important to note that the network was not trained on the tumor images nor fine-tuned, but was only used as a deep feature extractor. Moreover, the aim of this work is not to highlight the tumor or segment it by deep features, which could be addressed in future improvements, but only to explain how the network behaves while performing its diagnosis. The inconsistency in region-focusing represents an area for potential improvement, possibly through architectural modifications or additional training strategies tailored to enhance region-specific attention. Complementing the visual analysis, our framework provides a second layer of explainability through feature-level interpretation techniques that offer quantitative insights into the classification decision process. Fig. 6 illustrates the Feature Importance analysis, highlighting the most influential features

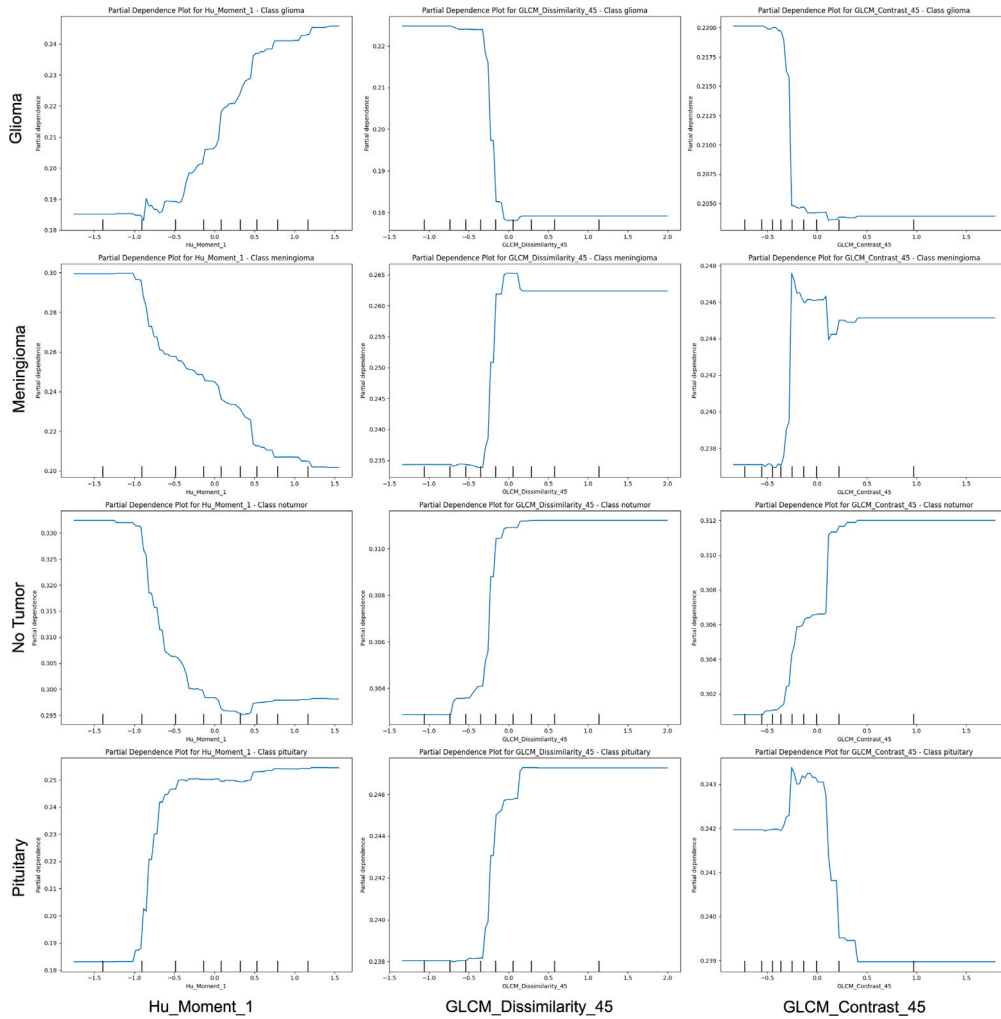


Fig. 7. Partial Dependence Plot about the three most influential hand-crafted features.

in the model's predictions. The features are ranked based on their impact on the model's accuracy, measured by the decrease in the accuracy score when each feature is randomly shuffled while the model-agnostic approach that ensures interpretability regardless of the underlying classifier.

Our analysis identified the most significant features as `Hu_Moment_1`, `GLCM_Dissimilarity_45`, and `GLCM_Contrast_45`, while other GLCM-based features and Hu moments also show moderate importance. Quantitatively, `Hu_Moment_1` has a median value of 0015 indicating that shuffling the first moment typically decreases the model's accuracies by approximately 1,5 percentage points. These findings are in line with domain knowledge that shape characteristics (captured by Hu moments) are critical diagnostic indicators for tumor identification. Some features exhibit high variability which reflects uncertainty in their importance, while others have values close to zero, suggesting that they contribute minimally to influence the final predictions. This analysis provides clinicians with a transparent understanding of which specific image characteristics most strongly influence classification decisions.

The final component of our explainability approach delves deeper into the relationship between specific features and class predictions, offering insights into how feature values affect diagnostic decisions. Fig. 7 presents PDP that illustrate the marginal effect of individual features on the predicted probability for each of the different tumor classes. Each subplot visualizes how changes in a specific feature influence the model's output while keeping other variables constant,

offering a deeper understanding of how each feature influences specific class predictions. These visualizations reveal non-linear relationships and threshold effects that would remain hidden in simpler analytical approaches, providing nuanced insights into the classifier's decision boundaries. The analysis reveals distinct patterns: `Hu_Moment_1` has a positive correlation with the probability of predicting glioma and pituitary tumors and demonstrates a negative correlation with meningioma and the absence of a tumor. From a clinical perspective, this can be interpreted as follows: higher values of the first Hu moment (which relates to the moment of inertia) are associated with the presence of glioma or pituitary tumors, while lower values suggest either meningioma or healthy brain tissue. This finding provides an objective, quantifiable characteristic that differentiates between tumor types. `GLCM_Dissimilarity_45` exhibits notable abrupt shifts, demonstrating that small changes in texture dissimilarity can trigger significant changes in classification probability. `GLCM_Contrast_45` also contributes significantly to the predictions for certain classes although its effects are less pronounced than the other key features.

In summary, our dual-level explainability approach provides complementary insights that enhance understanding of the model's decision-making process. The visual explanations offer intuitive, region-based interpretations that align with radiological practice, while the feature-level analyses provide quantitative evidence of feature relevance and their influence on diagnostic decisions. Together, these explanation techniques transform the "black box" nature of the classification process into a transparent, understandable system that can be effectively integrated into clinical workflows.

6.4. Limitations and future perspectives

While our framework demonstrates significant advancements in both classification performance and explainability, several limitations should be acknowledged and addressed in future research.

First, a more systematic feature selection methodology could potentially further optimize the model. The current framework utilizes all 27 hand-crafted features and 256 deep features without a formal feature selection process. Implementing techniques (such as recursive feature elimination or LASSO regularization) could identify the most discriminative subset of features, potentially improving both performance and computational efficiency.

The deep feature extraction process could be improved helping the model to achieve better performance and results interpretation, for example introducing the use of layers incorporating spatial attention mechanisms [48]. These attention mechanisms could enhance the model's focus on tumor regions, addressing the limitations observed in our Grad-CAM analysis where the network sometimes fails to precisely localize tumor boundaries.

Second, *BrAInVision* uses transfer learning on VGG16 for deep feature extraction without fine-tuning it or re-training it due to lack of computational power. In future developments, the training of a convolutional neural network from scratch on domain-specific 2D MRI images could enhance predictive performance by better capturing modality-specific features. Third, the model's capacity to generalize across different datasets requires further validation on multi-institutional data. While our cross-dataset validation provides initial evidence of generalizability, testing on more diverse datasets with varying acquisition parameters, scanner types, and patient demographics would provide stronger evidence of the framework's robustness in real-world clinical settings. Future research should focus on integrating multi-modal data from complementary imaging techniques (such as CT or PET scans) and exploring federated learning approaches to improve robustness and address privacy problems in medical data sharing. Moreover, future perspectives should include prospective validation in clinical environments confirming the framework's robustness, generalization qualities, and practical utility in real-world diagnostic practice.

7. Conclusion

The early and accurate diagnosis of brain tumors remains a critical challenge in modern neurology, with direct implications for patient survival. Despite significant advances in imaging techniques, the interpretation of MRI scans requires specialized expertise, is time-consuming, and can lead to diagnostic delays. AI-powered systems offer promising solutions, but their widespread adoption in clinical practice has been hindered by the "black box" nature of conventional approaches. This study introduced *BrAInVision*, a hybrid and explainable AI framework designed for brain tumor classification on MRI images. Our approach uniquely combines the predictive power of deep learning features with the interpretability of traditional hand-crafted features (Hu moments and GLCM), creating a balanced solution which ensures high level of explainability without sacrificing performance. *BrAInVision* demonstrated high performance across multiple evaluation scenarios, achieving an F1-score of 97% in the four-class classification task (Glioma/Meningioma/Pituitary/NoTumor) and 99% in binary classification, outperforming current state-of-the-art approaches. Importantly, these results were validated not only on the original test set but also on an independent validation dataset with expert-annotated tumor masks, confirming the framework's generalizability. The dual-level explainability approach represents a significant advancement for clinical adoption. Visual explanations allow radiologists to verify how neural network look at MRI images, while feature-level interpretations provide quantitative evidence of which imaging characteristics drive

the classification decisions. Together, these complementary explanations transform the diagnostic process from an opaque prediction to a transparent, verifiable analysis. *BrAInVision* is designed as a decision support system that augments radiological expertise rather than replacing it. By providing both accurate classifications and clear explanations for those classifications, our framework has the potential to enhance diagnostic confidence, reduce interpretation time, and ultimately improve patient outcomes through earlier and more accurate diagnoses. While our current implementation has demonstrated promising results, future research will focus on enhancing region-specific attention mechanisms, incorporating multi-modal imaging data, and conducting prospective clinical validation studies.

CRedit authorship contribution statement

Marco Gagliardi: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Danilo Mauro:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Tommaso Ruga:** Writing – original draft, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Eugenio Vocaturo:** Writing – original draft, Validation, Supervision, Methodology, Conceptualization. **Ester Zumpano:** Writing – original draft, Validation, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The paper contains all the data (see Section 5.1).

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